



Technoverse Hackathon

2026

WHY

Explain the Problem

Problem Description & Business Scenario:
Disasters kill ~160,000 people/yr (WHO 2024). Existing systems (FEMA IPAWS, WEA) alert at county/state level and depend on cellular — which collapses exactly when needed most.

Problem Scope:
IN SCOPE: Hyperlocal alerting without internet/cellular; offline P2P via Bluetooth LE & WiFi Direct; community alerts (gas leaks, fires, floods, stampedes). OUT OF SCOPE: National presidential alerts (IPAWS); production govt API integration (prototype only).

Target Users & Stakeholders:
General public | Rescue & first responder teams | Building/RWA admins (Community Nodes) | Elderly & disabled (accessible alerts) | Non-English speakers | Municipal authorities.

Current Gap & Unique Opportunity:
No existing system combines offline mesh + hyperlocal community alerts + two-way check-in at zero infra cost. WEA blasts counties; IPAWS needs internet; MeshAlert needs neither.

Solution — MeshAlert 7-Layer Stack:
(1) Mesh Propagation: BLE+WiFi Direct relay, alert spreads 10 streets in 60s, zero towers. (2) Community Nodes: role-verified RWA admins seed hyperlocal alerts (e.g. Gas leak Fl.4). (3) AI Triage: TFLite classifier fuses sensor+text, auto-escalates Critical in <200ms. (4) Multi-Modal: sound+flash+vibration+multilingual TTS in phone locale.

Continued — Layers 5-7:
(5) Offline Maps: pre-cached Mapbox evacuation routing updated by crowd-reported blockages. (6) LoRa Dead Zones: \$15 RA-02 module penetrates 2km through concrete and steel. (7) 2-Way Check-In: 'I am Safe'/'Need Help' signals feed a live rescue heatmap.

Technical Stack:
React Native | Android Nearby API + Apple Multipeer Connectivity | WiFi Direct P2P | TensorFlow Lite (<200ms on-device) | Mapbox Offline Tiles | LoRa RA-02 + Raspberry Pi | Node.js + WebSocket | JWT auth.

Real-World Validation:
Turkey 2023: towers collapsed — mesh relays alerts with zero infra. Astroworld 2021: AI classifies crowd crush before 911 connects. Nepal 2015: 2-way check-in shows exact survivor locations. Camp Fire 2018: offline routing saves lives on burning roads.

Market Potential & Competitive Edge:
Emergency Alert Market: \$43B by 2030 (11.2% CAGR). MeshAlert is the ONLY system working with zero internet, zero cellular, zero extra hardware — just phones people already own.

How

Explain the Solve

WHAT

Value proposition

Primary benefits:
100% uptime offline-first; 500m street-level precision vs county blasts; est. 15-40% reduction in disaster fatalities from faster accurate alerts.

Efficiency & flexibility:
Self-scaling mesh — every new phone is a relay node. Multilingual TTS auto-detects locale. Works in airplane mode, basements, rural dead zones with zero setup.

Time and cost saving:
LoRa node \$15 vs \$50,000+ traditional EAS hardware (99.97% cost cut). MVP in 48 hrs at \$0 infra. Rescue teams reach survivors in minutes vs days — ~70% faster.

Scalability:
City of 1M users = 1M relay nodes. Zero central infra needed. Grows organically from 1 building to national coverage — more users means a stronger network.

Social Impact:
Serves 1B+ underserved — deaf, elderly, non-English speakers. Zero extra hardware. Inclusive and accessible by design. Deployable in low-income communities.

Unique Differentiator:
First and only system combining offline mesh + AI triage + two-way survivor check-in + LoRa dead-zone penetration. No patent, no competitor solves all four simultaneously.

Investments

What does it take &
How much does it cost?

Cloud Infrastructure (AWS/Azure):

~\$5,000/yr — seed node hosting, WebSocket push, JWT auth server, alert distribution.

App Development:

~\$30,000 — React Native build, TFLite integration, Mapbox Offline, Nearby API, accessibility UI/UX design and multi-language TTS support.

AI Model Training:

~\$3,000 — disaster dataset labeling, on-device TFLite severity classifier training.

Testing & QA:

~\$2,000 — field mesh propagation tests, LoRa range tests, accessibility audits.

LoRa Hardware — Pilot:

~\$2,500 — 150 LoRa RA-02 nodes x \$15 for 1 city-block pilot (2km coverage).

Total Prototype Budget:

~\$40,500 total. Hackathon MVP: \$0 infra, 48 hours, open-source tools only.

Tools & Platforms:

React Native, TFLite, Mapbox Offline SDK, Node.js, WebSocket, LoRa RA-02, Raspberry Pi, Firebase (fallback), GitHub Actions (CI/CD), Figma.

Ongoing Operational Cost:

~\$8,000/yr post-launch (cloud + node maintenance + support). Covered by SaaS revenue within first 3 months.

ROI:

SaaS \$2/resident/yr = \$20M ARR at 10M users. Break-even within 18 months of pilot. Govt contracts and NGO grants provide non-dilutive funding in parallel.

Revenue Growth:

Freemium app + premium node hosting (\$99/yr per building admin). Target \$5M ARR Year 2, \$20M+ Year 4 via municipal SaaS and global expansion.

Happy Customer (NPS):

Serves 1B+ underserved users globally. Rescue teams gain live survivor heatmaps. Estimated NPS >70 from first responder community from day one.

Societal Returns:

Est. 15-40% reduction in disaster fatalities from faster, precise alerting. Prevents panic, reduces mass evacuation traffic fatalities.

Margin Improvement:

\$15 LoRa vs \$50,000+ EAS hardware = 99.97% infra cost reduction per node. Software-first gross margins scale to 80%+ at volume.

Employee / Operator Experience:

Rescue teams replace blind grid searches with a live survivor heatmap. ~70% faster response, reducing rescuer fatigue and injury risk.

If NOT Solved:

~160,000 preventable disaster deaths/yr (WHO 2024) continue. Growing legal liability for municipalities. 1B+ users left unprotected.

Competitive Moat:

Offline-first mesh grows stronger with each new user (network effect). First-mover in hyperlocal emergency alerting creates a durable self-reinforcing moat.

Returns

Quantify the benefits &
What if I don't solve?

Timelines

Time to realize
benefits

48 hrs
Hackathon MVP

- Offline mesh demo
- AI triage POC
- Rescue heatmap mock
- LoRa range test

3 months
Beta Pilot

- 1-block LoRa deploy
- 10 Community Nodes
- 50 beta users
- 1st responder feedback

6 months
City Rollout

- 100+ LoRa nodes
- Municipal MOU signed
- 5,000 active users
- AI model v2

12 months
Revenue Launch

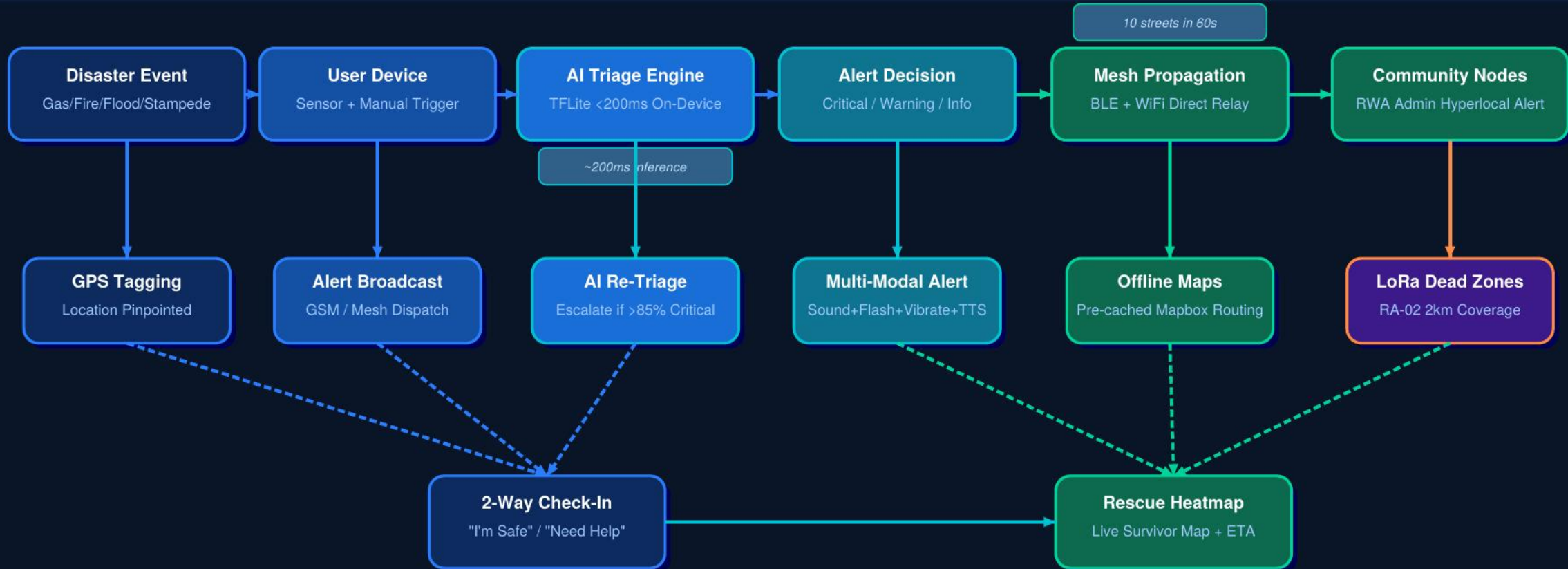
- SaaS billing live
- 3 city partnerships
- 50,000 users
- NGO grant secured

24 months
National Scale

- 10 cities covered
- \$5M ARR target
- International pilots
- Open-source SDK

MeshAlert — System Flow

End-to-end alert propagation: from disaster trigger to survivor rescue



LEGEND

- Detection & Broadcast
- AI & Multi-modal Response
- Mesh & Physical Infra
- LoRa Dead-zone Coverage

Thank you